Check for updates

Advanced Functional Materials

ADVANCED FUNCTIONAL MATERIALS
www.afm-journal.de

RESEARCH ARTICLE OPEN ACCESS

Near-Infrared Light-Driven Zn/Au Janus Micromotors for Multiplex SERS Detection of Anticancer Drugs

Tijana Maric[1,2] | Gohar Soufi[1,2] | Ziqiao Li[1,2] | Erick Chica[1,2] | Amran Abdirasheed Ali Jinow[1,2] | Nada Al-Roubaï[1,2] | Noor Ahmed[1,2] | Yaman Göksel[1,2] | Anja Boisen[1,2]

[1]The Danish National Research Foundation and Villum Foundation's Center for Intelligent Drug Delivery and Sensing Using Microcontainers and Nanomechanics, Department of Health Technology, Technical University of Denmark, Lyngby, Denmark | [2]Department of Health Technology, Technical University of Denmark, Lyngby, Denmark

Correspondence: Tijana Maric (tijma@dtu.dk) | Gohar Soufi (gohsoo@dtu.dk)

Received: 6 February 2026 | **Revised:** 22 April 2026 | **Accepted:** 8 May 2026

Keywords: anticancer drugs | micromotors | multiplex detection | NIR-light | SERS sensing | thermophoresis

ABSTRACT

Janus Zn/Au micromotors (MMs) are fabricated by sputtering Au onto one hemisphere of Zn microparticles, resulting in spherical light-responsive structures with dual photothermal propulsion and SERS detection capabilities. Zn/Au MMs exhibit efficient propulsion under near-infrared (NIR) light, with motion driven through thermophoretic effects induced by local heating of the Au surface. The MMs are activated by CARS microscopy, enabling label-free, real-time trajectory tracking, with propulsion velocities increasing linearly with total laser power and remaining largely independent of the pump wavelength within the NIR range (780–900 nm) under dual-wavelength excitation. A 30 nm Au coating provides the optimal balance between optical absorption and propulsion efficiency, while independent modulation of the pump and Stokes beams confirms that dual-beam excitation is essential for sustained motion. Zn/Au MMs work perfectly as active SERS platforms for single and multiplex detection of the anticancer drugs methotrexate (MTX) and 6-mercaptopurine (6-MP). Their active motion enhances analyte exchange at the Au surface, reducing affinity-driven saturation and competitive adsorption effects, which broadens the linear range and improves multiplex SERS detection. The integration of autonomous motion with chemometrics models, including partial least squares-discrimination analysis (PLS-DA) on the obtained SERS data, represents a promising step toward MM-based diagnostic systems.

1 | Introduction

Cancer is one of the leading causes of death worldwide. Most patients often receive combinations of two or more anticancer drugs to improve treatment outcomes. Finding precise technology for detecting anticancer drugs during therapy is highly important. Many anticancer drugs, including methotrexate (MTX) and 6-mercaptopurine (6-MP), must be dosed with high precision because their therapeutic range is very narrow. Even minor fluctuations in blood concentration can result in serious adverse effects or reduced therapeutic efficacy [1]. For this reason, real-time monitoring of drug concentrations in biological fluids (blood, urine, or saliva) would ensure both the safety and effectiveness of treatment [2, 3]. Furthermore, as combination chemotherapy becomes more common, the ability to measure several drugs at the same time is gaining importance [4]. Surface-enhanced Raman scattering (SERS) is a powerful, label-free technique that provides detailed molecular fingerprint information about drugs in a sensitive and selective manner [5, 6]. Its capability to identify and quantify multiple analytes simultaneously makes SERS valuable for the multiplex detection of drugs in complex biological environments [7, 8]. Plasmonic

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2026 The Author(s). *Advanced Functional Materials* published by Wiley-VCH GmbH

SERS substrates can enhance Raman signals by up to 10^{15}-fold, making them powerful for trace detection [9]. Among SERS-active materials, gold, silver, and copper nanostructures are the most widely used. Recent advances have focused on integrating these metals with high-surface-area materials, such as graphene and other carbon-based supports, to enhance analyte enrichment and detection [10–14]. Despite the SERS advantages (high sensitivity and selectivity, fast and real-time analysis, effective performance in complex biological environments), traditional plasmonic substrates are immobile and depend on passive analyte diffusion to the surface, which can reduce detection efficiency and reproducibility. Previously, researchers combined SERS technology with self-propelled micromotors (MMs), making efficient and dynamic systems for analyte detection and signal enhancement [15, 16]. Unlike static microparticles, MMs are microscale devices that can perform autonomous movement [17]. This motion can occur in various environments, spanning from simple aqueous solutions to complex biological media [18]. Their propulsion can be driven by external energy sources, such as magnetic fields, electric fields, light, or heat, but can also be triggered internally through chemical reactions between the MM and its surrounding medium [19–21]. Their ability to move comes from a process in which energy is turned into mechanical motion [22–25]. This self-driven mobility is one of the main advantages of MMs and offers clear benefits for SERS-based sensing approaches. Because they continuously move and redistribute molecules in their surroundings, MMs bring analytes into closer and more frequent contact with the SERS substrate. This gives stronger and more consistent signals and, overall, improves the quality of detection [26, 27]. In recent years, many researchers have turned to self-propelled SERS systems, aiming to pair active movement with plasmonic amplification to push detection sensitivity even further [28–30]. Li, Yang, and co-workers developed micromotors composed of magnetic beads coated with Ag nanoparticles and Au spikes (MB@Ag@Au MMs). The magnetic core allows external manipulation, while the Ag and Au components provide strong plasmonic activity [31]. When a magnetic field is applied, the MMs can be guided toward bacterial colonies or individual cells (*E. coli, S. aureus*). Their proximity to the cells and the strong plasmonic response make it possible to distinguish the two bacteria by their Raman signatures. The same magnetic-actuation principle was applied to develop rod-shaped Fe_3O_4@Ti_2C MMs. In this system, magnetic propulsion was integrated with charge-transfer-based SERS selectivity [32]. They efficiently concentrate analytes on MXene surfaces under rotating magnetic fields, resulting in up to ninefold stronger Raman signals and detection limits as low as 10^{-8} M for Rhodamine 6G (R6G). Remarkably, these MMs enable selective Raman enhancement by detecting only R6G, but not 4-mercaptopyridine, in mixed samples, due to the charge-transfer resonance between the MXene's Fermi level and the electronic states of the molecule. In contrast to magnetically actuated MMs, light has also been employed for propulsion through photopheresis. Escarpa, Sánchez, and their group used the same laser source for both triggering motion and detecting crystal violet (CV), achieving a 15–18-fold increase in the SERS signal after approximately 1000 s, as well as sensitive detection of malachite green (limit of detection (LOD) ~0.1 μM) and paraquat, even in salt-rich media. While MMs-SERS studies have demonstrated intracellular sensing, exosome targeting, and integrated photothermal theragnostic, direct detection of therapeutic drugs remains underrepresented, underscoring the need for motion-enhanced SERS platforms tailored to clinically relevant agents [33–36].

Herein, we fabricated Janus Zn/Au MMs that serve as dynamic SERS substrates for the detection of two anticancer drugs, 6-MP and MTX, both individually and simultaneously, exhibiting enhanced Raman intensity under motion. Compared to static systems, the active Zn/Au MMs produced stronger SERS signals and enabled dual-drug discrimination, with 6-MP exhibiting a higher affinity toward plasmonic hotspots, indicating motion-induced molecular enrichment and competitive binding effects. We showed that their motion comes from the thermophoresis mechanism under NIR light. The highest average velocities were observed for Zn microspheres coated with a 30 nm gold layer under simultaneous dual-wavelength irradiation, when the pump wavelength was set to 800 nm. They moved more quickly at higher laser intensities. To evaluate how they behave in a medium that resembles physiological conditions, we tested them in PBS, where they continued to move in a comparable manner.

2 | Results and Discussion

Figure 1 provides an overview of the fabrication of Janus Zn/Au micromotors, their movement, and multiplex drug detection using SERS. Commercial Zn microspheres were asymmetrically sputtered with Au to form Janus Zn/Au MMs with the thermal contrast needed for NIR-driven motion (Figure 1A). We tested several gold layer thicknesses to find the best balance between propulsion behavior and SERS activity. In the first part, motion studies were conducted under various laser wavelengths, powers, and Au thicknesses to establish the most effective propulsion parameters (Figure 1B). When exposed to NIR light, the Zn/Au MMs move due to the thermophoresis mechanism. In the second part of the study, we showed that they can be used as dynamic SERS substrates. Two anticancer drugs (6-MP and MTX) were tested as model drugs, both individually and together (Figure 1C).

The choice of material is guided by both propulsion performance and SERS functionality. Alternative material combinations, such as ZnO/Pt for catalytic propulsion or MB@Ag@Au MMs for SERS-based applications, have been explored. Pt MMs are widely used for catalytic propulsion, while Ag provides strong SERS enhancement. However, Pt lacks plasmonic activity in the near-infrared (NIR) range, making it unsuitable for SERS applications. Although Ag offers high SERS sensitivity, it exhibits lower photothermal efficiency under NIR irradiation. In contrast, Au combines strong plasmonic activity with efficient photothermal conversion, enabling dual functionality within a single platform.

The morphology and structure of the Zn/Au MMs were examined using scanning electron microscopy (SEM) to verify the successful Janus formation and the adsorption of drug molecules. As shown in Figure 2A, bare Zn microparticles exhibit uniform spherical morphology. After sputtering with Au, the Zn particles maintained their spherical morphology without noticeable surface disturbance or structural deformation (Figure 2B-i–iv). Energy-dispersive X-ray spectroscopy (EDS) elemental mapping confirmed the presence of both Zn and Au, verifying successful gold deposition on the particle surface (Figure 2B-ii, iii). The size distribution was determined by measuring 200 bare Zn

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

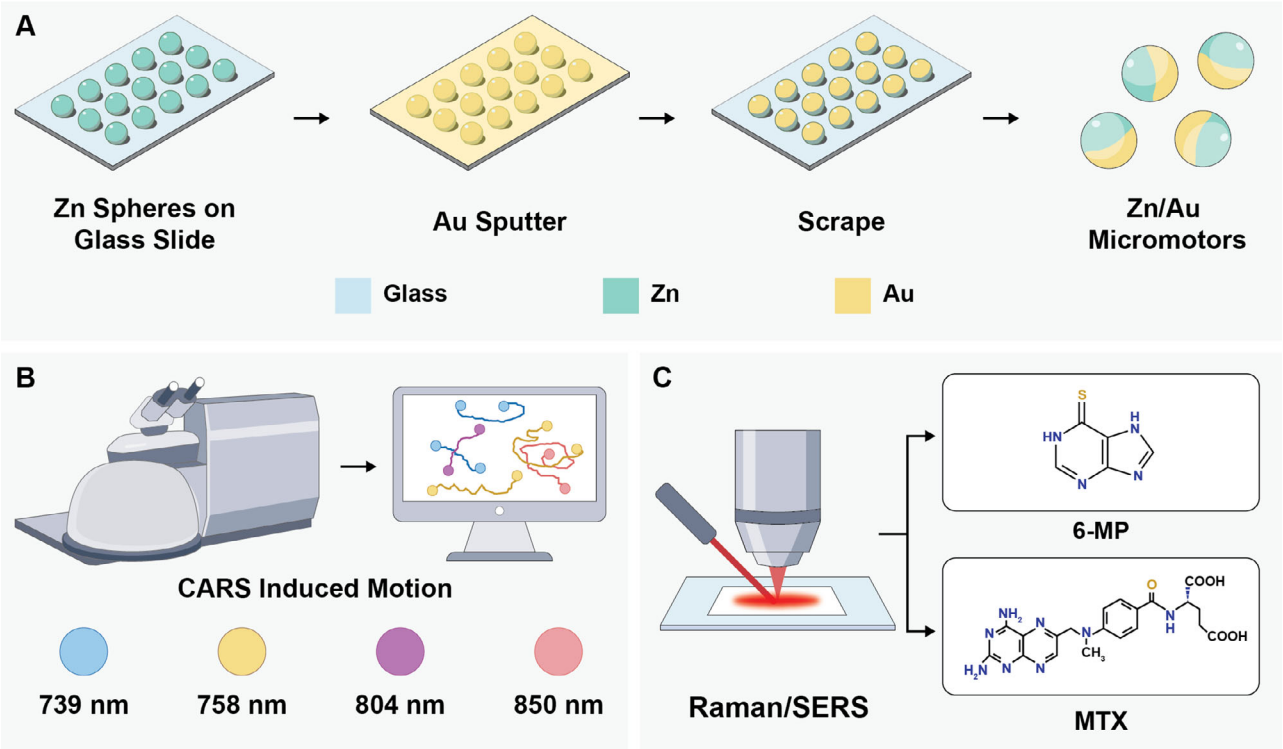


FIGURE 1 | Schematics of (A) fabrication procedure of Janus Zn/Au MMs; (B) motion study, and (C) SERS-based multiplex drug detection.

particles and 200 Zn/Au MMs from SEM images (Figure 2C, D, respectively). Figure 2C shows that most Zn particles have a size in the range of 2–3 μm, and that the gold coating did not significantly change the particle's diameter (Figure 2D). This uniformity in MMs' size is important, as the motion of MMs depends on their size. High variations in diameter could influence MMs' average velocity in subsequent motion studies.

Next, we examined the stability of the MMs stored in an aqueous solution and PBS for 15 days. It has been observed that there is a gradual degradation of MMs, which motivated us to systematically evaluate their structural stability and survival period after fabrication. SEM images were taken on days 1, 5, 9, and 15 after MMs preparation in aqueous solution (Figure 2E) and PBS (Figure 2F) and accounted for the total number of MMs in a 4 μL sample on days 1 and 15 (Figure 2G). The number of MMs counted on day 1 was defined as 100% ($n = 3$), and the percentage of MMs survived until day 15 was calculated accordingly. Figure 2G shows an 81% decrease in the number of MMs after 15 days of storage in aqueous solution and 88.8% decrease after 15 days of storage in PBS. These results indicate MMs' degradation. SEM images presented in Figure 2E,F confirmed MMs' surface corrosion by day 15 with more pronounced degradation and thus faster corrosion in PBS. Furthermore, extended degradation studies showed that Zn/Au MMs undergo complete degradation and fully disappear after approximately 20 days in both aqueous solution and PBS.

This degradation occurred not only on the Au layer but also in the Zn core. It can be explained by the different electrochemical reactivity of the two metals. Zn acts as a sacrificial anode in the galvanic couple as it is more reactive and prone to oxidation than Zn. Zn loses electrons and dissolves over time, while Au remains relatively inert due to its low reactivity and resistance to oxidation. The apparent loss of MMs within 15 days indicates that Zn/Au MMs are unstable in aqueous solution or PBS. Higher MMs degradation in PBS could be explained by the presence of ionic species, which increases conductivity and promotes galvanic corrosion. Most likely, Cl^- ions destabilize the protective layers on the Zn surface, making them more porous and breaking down the passivation layer.

A similar trend was previously observed by Pumera et al. for Pt-based MMs. They proved that even noble-metal MMs are prone to corrosion over time [37, 38]. A similar degradation behavior has been described for transient MMs made from Zn and Mg, where the presence of two different metals leads to galvanic effects that speed up their dissolution in both water-based and acidic environments [39]. Our study, along with other reports, highlights that MMs must be used right after their preparation to ensure they perform at their highest motion efficiency. To avoid corrosion, Zn/Au MMs could also be stored as dry powders.

2.1 | Propulsion Behavior of Zn/Au MMs

First, we studied the motion behavior of Zn/Au MMs under NIR light in water media and then in PBS. This matters because how fast they move has a clear influence on how much they can enhance the SERS signal. We varied a few parameters: laser wavelength, Au thickness, and laser power, to study how each of them affected the MMs velocity. For these experiments, a Coherent Anti-Stokes Raman Scattering (CARS) microscope has been used as a light source. A CARS setup uses two lasers, a Pump laser with a tunable wavelength and a Stokes laser fixed at 1031 nm. We used both lasers simultaneously to start the motion

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

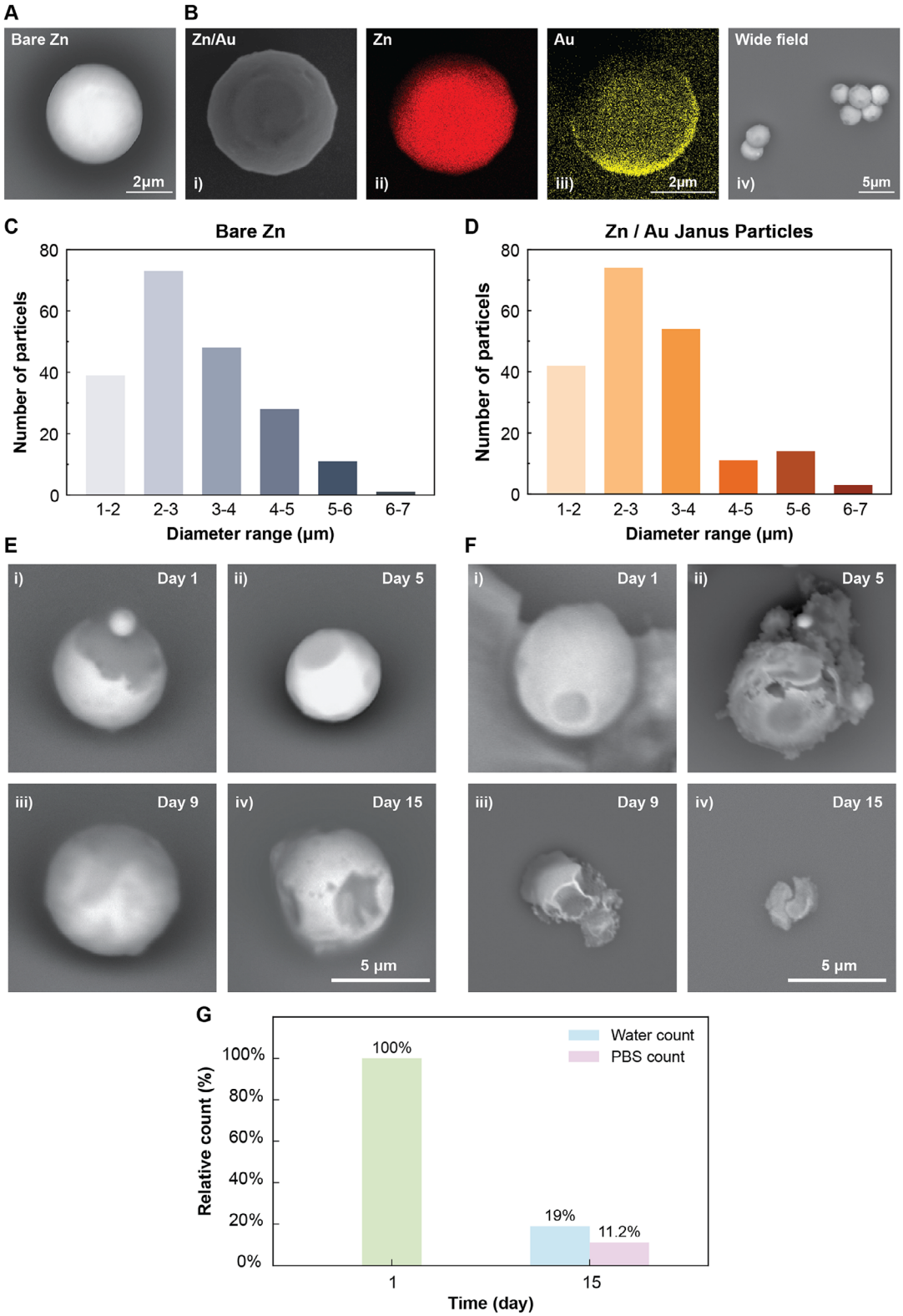


FIGURE 2 | Scanning electron microscopy (SEM) images of (A) bare Zn particles, (B) Zn/Au MMs: (i) SEM image of a single particle, (ii,iii) corresponding energy-dispersive X-ray spectroscopy (EDS) elemental mapping (Zn in red, Au in green), and (iv) SEM image showing multiple MMs (wide field). Size distribution of (C) bare Zn, and (D) Zn/Au MMs, obtained by measuring 200 MMs from SEM images. (E) SEM images of the gradual degradation of Zn/Au MMs over 1, 5, 9, and 15 days in aqueous solution. (F) SEM images showing the degradation of Zn/Au MMs in PBS over the same time (1, 5, 9, and 15 days). (G) Quantitative analysis of MM degradation, expressed as the percentage of remaining MMs on day 15 relative to day 1 (defined as 100%, n = 3 independent measurements) in aqueous solution and PBS.

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of the MMs. The frequency difference between the lasers aligns with vibrational modes in the sample, enabling NIR excitation and imaging of the MMs [40]. The CARS system allows us to adjust the wavelength, power, and detectors, giving better control over the MMs' motion and their interaction with the light [41].

In the first set of experiments, we tested the motion of the MMs at different Pump laser wavelengths. The Stokes laser wavelength was kept constant at 1031 nm throughout all experiments. MMs were simultaneously irradiated with both the Pump and Stokes beams, while the pump laser wavelength was varied between 780, 800, 850, and 900 nm. Figure S1 shows the average velocities of 30 nm Au-coated MMs measured under simultaneous dual-wavelength irradiation at the different Pump wavelengths. The total laser power from both beams was maintained at 17 mW. The corresponding average velocities were 9.46 ± 2.23, 9.32 ± 1.86, 8.78 ± 2.41, and 8.09 ± 2.27 μm s^{-1}, respectively. The two-tailed t-tests showed no significant differences between the wavelengths ($p > 0.05$). The results show that the velocity of the MMs does not depend on the chosen Pump laser wavelength and that they can move over a broad NIR range. Our earlier study also showed that the MMs exhibited similar velocities regardless of the NIR excitation wavelength [41]. This indicates that the MMs are propelled not by plasmon resonance but by a photothermal effect. This effect depends on how efficiently the Au layer absorbs light, not on any specific resonant wavelength. The Au part of the MMs is responsible for absorbing the light and converting it into heat [42]. The photothermal behavior of the Zn/Au MMs was evaluated by monitoring the temperature increase under 808 nm irradiation at a power density of ~1.27 W cm^{-2}, with the corresponding heating and cooling profiles shown in Figure S2a and the linear fitting of ln θ versus time during the cooling process presented in Figure S2b. This generates localized heating and a temperature gradient around the MMs. The Au-coated side becomes hotter than the bare Zn hemisphere. Consequently, the generated thermophoretic force drives MMs via a thermophoretic mechanism [43]. The photothermal conversion efficiency (η) of the 30 nm Au-coated MMs was calculated to be approximately 47.3%, indicating efficient light-to-heat conversion (see Supporting Information for details). Since the velocities were similar with no significant statistical differences at all studied Pump wavelengths, for all following experiments, the pump laser wavelength was fixed at 800 nm. The second parameter investigated was the thickness of the coated Au layer. Figure 3A shows the average velocities of the Zn/Au MMs for Au coating thicknesses of 15, 30, 37, and 45 nm. The experiments were conducted at a laser power of 19.2 mW. Under these conditions, the Zn/Au MMs coated with 15 nm of Au did not display any clear propulsion. This is likely due to the poor photothermal efficiency of very thin Au layers (<20 nm). When the Au layer is that thin, it does not spread into a smooth film but instead breaks into small islands on the curved Zn surface. As a result, it does not absorb NIR light very well, and the heating it produces is limited. Because the propulsion of the MMs under NIR illumination depends on a photothermal-induced thermophoretic mechanism, the inadequate thermal gradient at this coating thickness prevents measurable motion. By contrast, increasing the Au coating thickness resulted in measurable propulsion of the Zn/Au MMs. Figure 3A shows that there is a statistically significant difference in average velocity between the Zn/Au MMs with an Au thickness of 30 nm (8.98 ± 1.50 μm s^{-1}) and 37 nm (5.43 ± 0.84 μm s^{-1}) at a laser power of 19.2 mW ($p < 0.05$). Although an increase in the Au coating thickness is expected to enhance NIR light absorption and thereby improve photothermal energy conversion, a thicker Au layer also increases the overall mass of the MMs, making them heavier and more difficult to propel via the thermophoretic mechanism at the same laser power. Consequently, higher energy input (stronger laser power) would be required to achieve comparable propulsion velocities. The Zn/Au MMs coated with a 45 nm thick Au layer did not exhibit any propulsion under 19.2 mW NIR illumination. This can be attributed to reduced photothermal efficiency at higher coating thicknesses. When the Au layer becomes optically thick, additional gold mainly increases reflectivity and enhances lateral heat dissipation rather than improving light absorption. As a result, the generated temperature gradient in the surrounding medium becomes insufficient to induce a thermophoretic force, and the heavier MMs remain immobile. For clarity, a summary of the motion analysis under different experimental conditions is provided in Table S1.

Based on the previously observed influence of Au coating thickness on the propulsion behavior of Zn/Au MMs at low laser power, the next step was to investigate whether increasing the laser power could enhance their motion. We aimed to determine whether a higher irradiation power could induce propulsion in MMs with 15 and 45 nm Au coatings, which showed no movement at 19.2 mW, and whether the same trend with Au thickness would persist under stronger illumination. For this purpose, the laser power was increased to 43.8 mW, and the average velocity was measured for the same four Au thicknesses (Figure 3B). Even with increased laser power, Zn/Au MMs coated with 15 nm Au remained immobile, consistent with the previously observed behavior at lower power, where the discontinuous morphology of the ultrathin Au layer limited its photothermal heating efficiency. For the MMs with the thicker Au coatings, the propulsion behavior changed when we increased the laser power to 43.8 mW. The MMs coated with a 45 nm Au layer, which didn't move at the lower power, started to propel. MMs coated with 30 and 37 nm Au thickness also showed an increase in velocity. The average velocities reached 15.34 ± 3.75 μm s^{-1} for 30 nm coatings, 15.27 ± 3.62 μm s^{-1} for 37 nm coatings, and 19.82 ± 3.81 μm s^{-1} for 45 nm coatings. They differed significantly between the 30 and 45 nm Au-coated MMs ($p = 0.000788$) and between the 37 and 45 nm Au-coated MMs ($p = 0.000558$). No significant difference in average velocity was observed between the 30 and 37 nm Au-coated MMs ($p = 0.952751$). When we increased the laser power from 19.2 to 43.8 mW, the 30 nm Au-coated MMs showed only a slight increase in velocity ($\Delta v = 6.36$ μm s^{-1}). The 37 nm Au-coated MMs responded with a much larger difference in velocity ($\Delta v = 9.84$ μm s^{-1}). This thickness requires more energy to produce a more substantial photothermal effect. The 45 nm Au-coated MMs, however, gained enough energy at the higher laser power to compensate for their lower absorption efficiency. As a result, they effectively generated a local thermal gradient and showed the highest average velocity. These results suggest that MMs with thinner Au layers can produce sufficient heat even at lower laser powers, whereas those with thicker coatings need more energy to reach similar propulsion levels.

After confirming that the propulsion of the Zn/Au MMs depends on both the Au coating thickness and the laser power, we carried out the remaining experiments using a fixed setup: a Pump

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

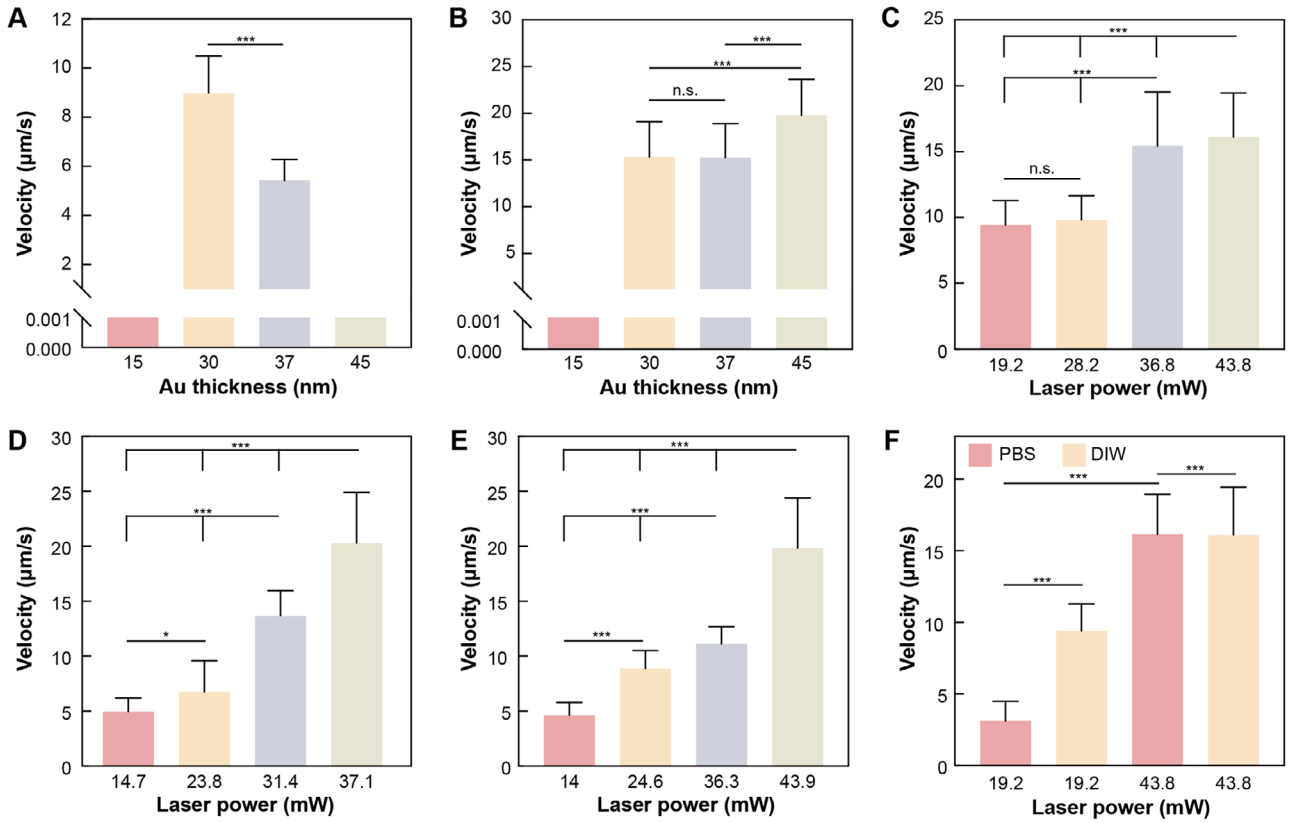


FIGURE 3 | (A) Average velocities of Zn/Au MMs with Au coatings of 15, 30, 37, and 45 nm under dual-beam NIR excitation (total power = 19.2 mW). (B) Effect of increased laser power (43.8 mW) on the average velocity of Zn/Au MMs with different Au coating thicknesses. (C) Dependence of average velocity on total laser power for Zn/Au MMs (30 nm Au coating) under balanced dual-beam excitation. (D) Average velocity of Zn/Au MMs (30 nm Au coating) as a function of Stokes laser power with constant Pump power (20%). (E) Average velocity of Zn/Au MMs (30 nm Au coating) as a function of Pump laser power with constant Stokes power (20%). (F) Comparison of average velocities of Zn/Au MMs (30 nm Au coating) in DI water and PBS at 19.2 and 43.8 mW total power. All experiments were performed under dual-beam NIR excitation with the Pump laser fixed at 800 nm and the Stokes laser at 1031 nm. For all experiments, the Pump laser wavelength was fixed at 800 nm and the Stokes laser at 1031 nm. Velocities were calculated from trajectory analysis of individual MMs ($n = 20$ MMs per condition). Data are presented as mean ± standard deviation (SD). Statistical significance was evaluated using a two-tailed t-test, with $p < 0.05$ considered statistically significant.

wavelength of 800 and a 30 nm Au coating. This configuration provided stable propulsion under all tested conditions. In the following experiments, we adjusted the Pump and Stokes laser powers to see how their combined effect influences MM propulsion. We increased the Pump and Stokes powers in parallel to 20%, 30%, 40%, and 50% of their maximum outputs, resulting in total measured powers of 19.2, 28.2, 36.8, and 43.8 mW, respectively (Figure 3C). As shown in Figure 3C, the average velocity increased with an increase in laser power. At 19.2 mW, the MMs reached an average velocity of 9.42 ± 1.86 μm s^{-1}. By increasing laser power to 28.2 mW, the average velocity slightly increased to 9.83 ± 1.82 μm s^{-1}. A higher jump in average velocity was observed at 36.8 mW (15.45 ± 4.07 μm s^{-1}), and the highest average velocity was obtained at 43.8 mW (16.11 ± 3.33 μm s^{-1}). The smaller power increments showed no statistical difference in velocity (19.2 vs. 28.2 mW and 36.8 vs. 43.8 mW), whereas the jump between larger power levels resulted in a statistically significant change in velocities (28.2 vs. 36.8 mW, p = 0.00000436). The results suggest that MMs' acceleration does not respond linearly to light. They require a minimum laser power for photothermal heating to become effective. When the power is too low, the temperature gradient around the MMs is insufficient to generate thermophoretic motion. Once this threshold is crossed, the absorbed light produces significantly higher thermal gradients, and MMs begin to move. This power setting served as a reference point for the rest of the study. In the following experiments, we varied the power of only one laser while keeping the other fixed to determine whether and how each contributes to MM propulsion.

In the next set of experiments, we explored how the Zn/Au MMs with a 30 nm Au-coating respond to variations in the Stokes laser power. The Pump power was held at 20%, while the Stokes was gradually increased from 10% to 30%, 50%, and 70% of its maximum output. These settings resulted in total powers of 14.7, 23.8, 31.4, and 37.1 mW, respectively. As shown in Figure 3D, the average velocity of the Zn/Au MMs increased with increasing Stokes laser power, following an approximately linear trend. The average velocity at the lowest power (14.7 mW) was 4.97 ± 1.21 μm s^{-1} and increased to 6.75 ± 2.82, 13.68 ± 2.29, and 20.28 ± 4.62 μm s^{-1} at 23.8, 31.4, and 37.1 mW, respectively. Statistical analysis showed that all comparisons between the measured average velocities resulted in $p < 0.05$, meaning that even small increases in Stokes laser power had a significant impact on propulsion. This shows that the Stokes laser adds distinctly to the photothermal heating of the Au surface by boosting the temperature gradient that enables thermophoretic motion.

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

In a similar setup, the same approach was applied, but this time the power of the Pump laser was varied from 10% to 30%, 50%, and 70% of its maximum output, while the power of the Stoke laser remained constant at 20%. The total measured powers were 14, 24.6, 34.3, and 43.9 mW, respectively. As shown in Figure 3E, the average velocity of the Zn/Au MMs increased with an increase in the Pump laser power. The average velocity at 14 mW was 4.62 ± 1.16 μm s^{-1}, rising to 8.89 ± 1.61, 11.12 ± 1.55, and 19.84 ± 4.53 μm s^{-1} at 24.6; 34.3; and 43.9 mW, respectively. Statistical analysis showed that all differences in velocity were statistically significant.

Overall, the MMs' propulsion depends on the combined power input from the Pump and Stoke lasers. The Stoke laser contributes more to the photothermal response, as it produces higher velocities at lower total power. The Zn/Au MMs moved only when the Pump and Stokes lasers were switched on simultaneously. Switching off either laser immediately stopped the motion. Thus, the combined input of both beams is required to generate the photothermal gradient that drives MMs in the CARS setup. Neither beam alone generates enough localized heating to maintain a stable temperature gradient across the Janus interface; only their combined absorption enables thermophoretic propulsion.

To test how the Zn/Au MMs behave in a more realistic ionic environment, we replaced water with PBS, a physiological buffer. This step is an essential part for determining whether their propulsion can be maintained beyond simple water-based systems. As shown in Figure 3F, Zn/Au MMs with a 30 nm Au-coating can move in both media under NIR illumination at 19.2 and 43.8 mW power, with corresponding representative motion shown in Videos S1–S4. Corresponding trajectory and mean square displacement (MSD) analyses are provided in Figure S3. The MSD increases with time, indicating active, non-Brownian motion of the MMs under the studied conditions.

At the laser power of 19.2 mW, the average velocity in PBS media was 3.12 ± 1.37 μm s^{-1} compared to the average velocity in DI water, which was much higher, 9.42 ± 1.86 μm s^{-1} ($p = 2.65 \times 10^{-14}$). It was expected, as they have different viscosities (1.3 mPa·s for PBS and 0.93 mPa·s for DI water) due to the presence of dissolved ions in PBS that can influence thermophoretic forces. However, when the laser power increased to 43.8 mW, the Zn/Au MMs achieved similar velocities in both media (16.11 ± 3.33 μm s^{-1} in DI water and 16.17 ± 2.76 μm s^{-1} in PBS).

As the laser power increases, the Au portion of the MMs heats up more. It lowers the local viscosity of PBS and strengthens the temperature gradient across MM. As a result, the MM can move more easily despite the ionic and viscous drag, reaching velocities comparable to those in DI water. The results indicate that Zn/Au MMs remain capable of efficient propulsion in physiologically relevant environments, which is an encouraging step toward real biomedical applications. Control experiments under visible-light illumination (405–600 nm) produced no measurable motion, confirming that the propulsion of Zn/Au MMs originates from NIR-induced photothermal effects.

2.2 | SERS-based Detection of Anticancer Drugs

The use of light-powered Zn/Au MMs provides an additional degree of freedom for label-free SERS-based detection by enabling remote and controllable manipulation of substrate motion. The Au layer served as both a plasmonic component for SERS enhancement and a protective shell improving the chemical stability of the Zn cores. In this work, light-powered Zn/Au MMs with an Au shell thickness of 30 nm were employed, enabling efficient electromagnetic enhancement and propulsion.

MTX and 6-MP have been used as model analytes with distinct molecular properties. No detectable Raman signal was observed in the absence of MMs, confirming that the analyte concentration alone was insufficient for spontaneous Raman detection (Figure 4A,B). Upon introduction of the Zn/Au MMs, characteristic Raman peaks of MTX (670 and 1365 cm^{-1}) and 6-MP (440 and 1220 cm^{-1}) were clearly resolved, demonstrating the intrinsic SERS activity of the Zn/Au MMs substrates (Figure 4A,B). The Raman bands at 670 cm^{-1} (MTX) and 440 cm^{-1} (6-MP) were selected for analysis due to their spectral specificity, minimal overlap between analytes, and stable intensity across the investigated concentration range, despite the presence of more intense but overlapping peaks. In addition, we systematically investigate the effect of Au shell (15, 30, 37, and 45 nm) thickness on the MTX Raman shift (Figure S4) and demonstrate that the observed signal enhancement is strongly correlated with the propulsion speed of the micromotors (Figure 3A) and also with UV–vis spectra of MMs (Figure S5). This behavior can be attributed to a combined effect of plasmonic enhancement and propulsion-driven mass transport, where Au thickness influences both the electromagnetic field enhancement and the micromotor speed. To evaluate the broader applicability of the platform, additional compounds were investigated, including trans-1,2-bis(4-pyridyl)ethylene (BPE) as a model Raman-active molecule and 5-fluorouracil (5-FU) as other types anticancer drug with a distinctly different molecular structure and size compared to MTX and 6-MP. As shown in Figure S6, clear and characteristic Raman peaks of 5-FU, the band at ~785 cm^{-1}, which is attributed to the F-C band, were observed [44]. These results demonstrate the capability of the Zn/Au Janus MMs to detect structurally diverse analytes, highlighting the versatility of the system. The enhancement in non-activated MMs arises from localized electromagnetic-field amplification at the Au surface. It is considered a passive SERS effect, in which signals depend solely on analyte diffusion and random contact with the substrate. Notably, when MMs were set in motion by light, a substantial increase in Raman intensity was observed at low MTX concentrations and high 6-MP concentrations (Figure 4C,D). This enhancement can be attributed to several synergistic effects: (i) the micromotors actively move analyte molecules to SERS-active areas, (ii) increased local analyte concentration due to the transitory and continuously renewed localization of analytes, and (iii) formation of temporary electromagnetic "hotspots" arising from MMs clustering. Together, these effects make the interaction between the substrate and analyte much more efficient than in static conditions.

The effect of a motion-based system is altered for analytes with different properties and affinities for the SERS substrate surface.

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

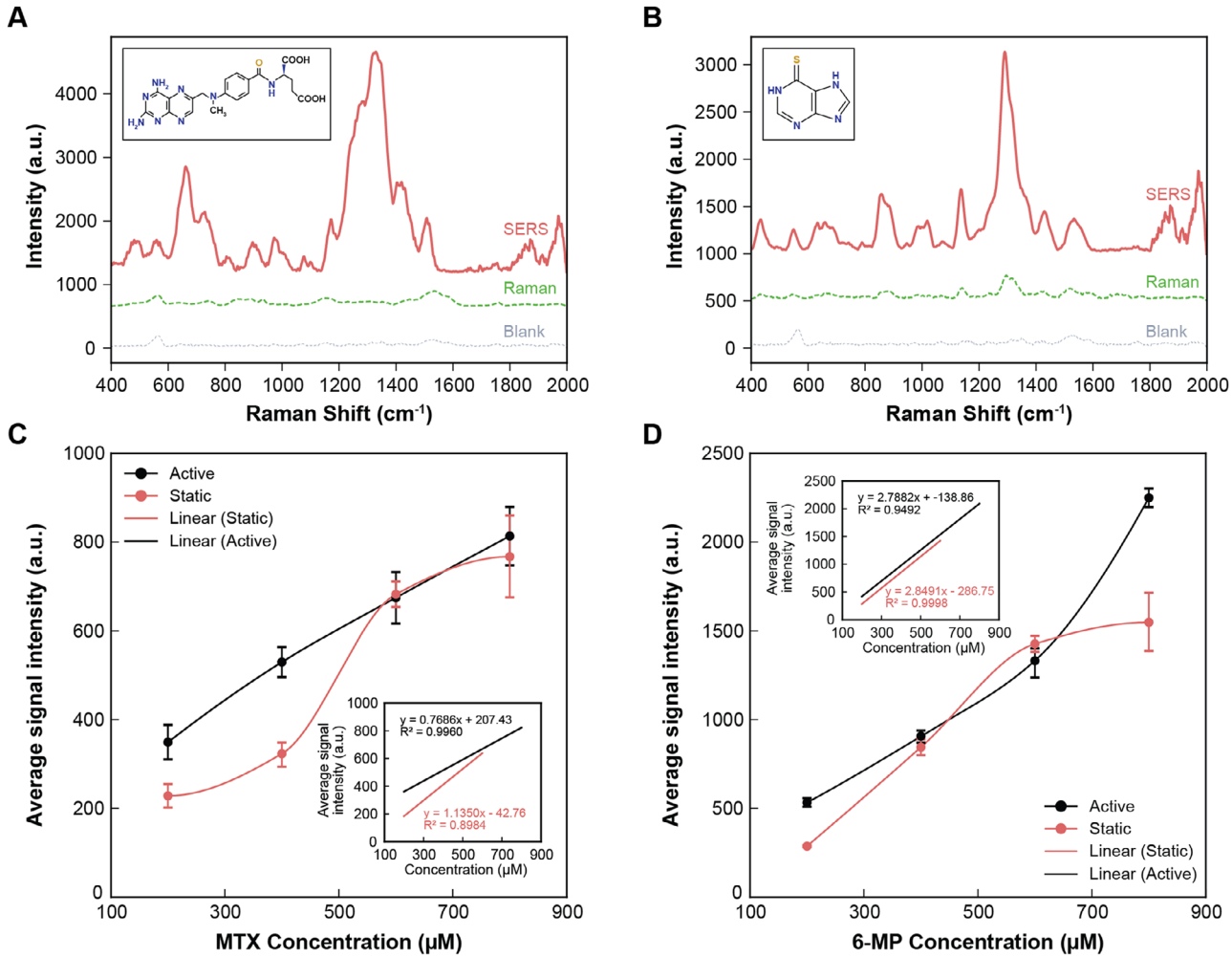


FIGURE 4 | Raman and SERS spectra of (A) MTX and (B) 6-MP on the bare substrates and Zn/Au MMs at 600 μM concentrations. (C) Average signal intensity of 670 cm^{-1} versus concentration curve for MTX using active and static Zn/Au MMs at concentrations of 200, 400, 600, and 800 μM. The black line represents active Zn/Au MMs, and the red line represents static Zn/Au MMs. A linear function was used to fit the SERS data. (D) Average signal intensity of 440 cm^{-1} versus concentration curve for 6-MP using active and static Zn/Au MMs at concentrations of 200, 400, 600, and 800 μM. The black line represents active Zn/Au MMs, and the red line represents static Zn/Au MMs. A linear function was used to fit the SERS data (inserts in the C and D). Data are presented as mean ± standard deviation (SD) obtained from three independent Raman mappings per condition.

For the drug molecule containing a thiol functional group (6-MP), which exhibits a strong affinity for gold surfaces, the passive SERS mode rapidly reached saturation at higher concentrations. This behavior is consistent with surface site saturation, where analyte molecules densely occupy available SERS-active sites, limiting further signal increase. In contrast, under active measurement conditions, the Zn/Au MMs' dynamic motion continuously refreshed the interaction environment, mitigating local saturation effects. As a result, a significantly extended linear detection range was achieved, highlighting the ability of active MMs to overcome affinity-driven signal plateauing [45].

For the MTX, characterized by weaker substrate affinity in comparison with 6-MP, the benefits of active mode are not only extending the linear range but also higher signal intensity at low concentrations (Figure 4C). Under light-driven conditions, Raman intensities for low concentrations of MTX showed an average of 50% enhancement compared with passive measurements, which is due to an increase in the collision frequency and analyte capture efficiency. This observation underscores the importance of active transport in compensating for low binding probabilities and diffusion-limited interactions, which are common challenges at low analyte concentrations. The limit of detection (LOD) and limit of quantification (LOQ) were estimated from the calibration data based on the blank signal. For 6-MP, the LOD and LOQ were approximately 4.5 and 13.5 μM, respectively, while for MTX they were approximately 14.5 and 45 μM in single detection mode. In the next step, we investigate the changes in the SERS signal under multiplex detection conditions and the influence of competitive interactions between analytes. These values indicate that, although the calibration range focuses on higher concentrations, the platform is capable of detecting analytes at lower micromolar levels. In high-dose MTX therapy, serum concentrations can reach the high micromolar range (hundreds up to ~1000 μM) during or shortly after infusion, and subsequently decrease to approximately 50 μM up to 24 h, highlighting that the investigated concentration range reflects clinically relevant early-phase exposure conditions [46]. It should be noted that these performance metrics were obtained in PBS and deionized water. In complex biological matrices, matrix effects such as protein interference

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

may impact detection sensitivity. In our previous work, such challenges have been addressed through appropriate sample pretreatment strategies in combination with established SERS substrates and validation against reference analytical methods. These results suggest a feasible pathway for future adaptation and validation of the present system in biologically relevant conditions [47].

2.3 | Multiplex Detection

Beyond individual analyte measurements, multiplex detection was further investigated to show the difference between the active and passive Zn/Au MMs. The benefits of active MM-assisted SERS become obvious in multiplex detection, where analytes compete for a finite number of SERS-active sites [45]. In the passive detection mode, the presence of a high-affinity analyte leads to preferential adsorption onto the Zn/Au surface, effectively blocking available enhancement sites. As a result, the Raman signal of the MTX, which exhibits lower Au surface affinity, is significantly suppressed due to competitive binding and surface saturation effects specifically in high concentrations of 6-MP (Figure 5A–C). This competition-induced signal reduction limits the reliability of passive SERS for simultaneous detection of multiple analytes.

In contrast, under active detection conditions, light-powered Zn/Au MMs continuously redistribute the SERS substrate and dynamically refresh drug-substrate interactions. The active transport and transient aggregation of MMs increase the effective number of available SERS-active sites and reduce prolonged occupation by high-affinity molecules. Consequently, competition between analytes is lessened, leading to a markedly smaller reduction in the Raman signal of the second, lower-affinity molecule (Figure 5B–D). This behavior indicates that active MMs motion mitigates surface site monopolization by dominant species and enables more balanced signal contributions in multiplex measurements. For further investigation, the identification and discrimination of two drugs and their mixture, we have implemented a PLS-DA model using preprocessed SERS spectra. The data sets consist of nine points with MTX concentration at 600 and 800 μM, five points of 6-MP at 600 μM, and five points of a mixture of them. The loading plots were examined to identify the main Raman shifts that make discrimination between MTX, 6-MP, and their mixture. As it is observable in Figure 5E, the dominant Raman Shifts in LV1 are 670 and 1360^{-1}, which are mainly the Raman Shifts attributed to the MTX [46], and we have also minor positive contributions of 440 and 1220 cm^{-1}, which are the Raman Shifts of 6-MP. In contrast, in LV2, the positive contributions are coming from 670 cm^{-1}, and the negative contribution is coming from 440 cm^{-1}, which are characteristic peaks of MTX and 6-MP, respectively. The score plot indicates that the main discrimination between MTX and 6MP originates from LV2, which yields positive scores for MTX and negative scores for 6-MP (Figure 5F). The calibration results indicated perfect classification, with all the samples correctly assigned to their respective classes. Sensitivity, specificity, and precision were 1.00 for all classes. The confusion matrix confirmed complete separation, with no misclassified or unassigned samples. Model robustness was evaluated using cross-validation, which resulted in only one misclassification: a mixture sample predicted as 6-MP. This yielded an overall predictive accuracy of 95.5%. Thus, the PLS-DA model enables clear multiplex detection and identification of both targeted drugs by active Zn/Au MMs.

Collectively, these results demonstrate that light-powered MMs amplify SERS signals for the weakly interacting analytes and they fundamentally alter the detection dynamics. By enabling active analyte enrichment and mitigating surface saturation effects, the approach broadens the linear dynamic range and enhances sensitivity, particularly for low-concentration or weakly interacting analytes. This dynamic SERS strategy is therefore highly promising for practical applications in drug monitoring and complex bioanalytical environments, where both sensitivity and quantitative reliability are critical.

3 | Conclusions

In summary, this work demonstrates the fabrication of Janus Zn/Au MMs that couple light-driven propulsion in the NIR region with strong SERS activity. Zn/Au MMs exhibit tunable propulsion behavior, governed by the Au coating thickness and laser power, with a 30 nm configuration identified as optimal. Their motion remains stable in ionic media such as PBS, confirming robustness under physiologically relevant conditions. The light-propelled Zn/Au MMs act as active SERS platforms for single and multiplex detection of the anticancer drugs MTX and 6-MP. In the case of single drug detection, the MMs enhance analyte diffusion and enrichment. The continuous renewal of interaction sites results in expanding the linear dynamic range for the drug with high affinity to substrates (6-MP), by mitigating surface saturation effects. Also, boosts sensitivity, particularly for low concentrations in the case of weakly interacting analytes (MTX). For multiplex detection, competitive adsorption effects between MTX and 6-MP were observed, underscoring the complexity of multiplex sensing at nanoscale interfaces. Chemometric analysis using PLS-DA enables reliable classification under multiplex conditions, and future work will explore more advanced machine learning approaches for quantitative analysis of complex Label-free SERS datasets. Overall, active detection not only broadens the linear range and sensitivity but also improves multiplex detection performance by minimizing competitive adsorption effects. Thereby, active detection enables more accurate and reliable simultaneous detection of analytes with different surface affinities using their SERS fingerprint. These results highlight Zn/Au MMs as dual-functional systems that integrate autonomous motion and SERS detection, paving the way toward intelligent MM-based diagnostic platforms.

4 | Experimental Section

4.1 | Fabrication of Zn/Au MMs

Zn particles (Sigma–Aldrich) were used as a template for MM fabrication. 1200 mg of Zn particles were dispersed in 35 mL of ethanol (99.8%). Subsequently, it was washed three times with ethanol, followed by sonication for 10 min at 25°C (Branson 2210 DTH Ultrasonic Cleaner, Branson Ultrasonics, USA) and centrifugation at 3000 rpm for 2 min (ST16R centrifuge, Thermo Scientific). After each step, the sample was replaced with fresh

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

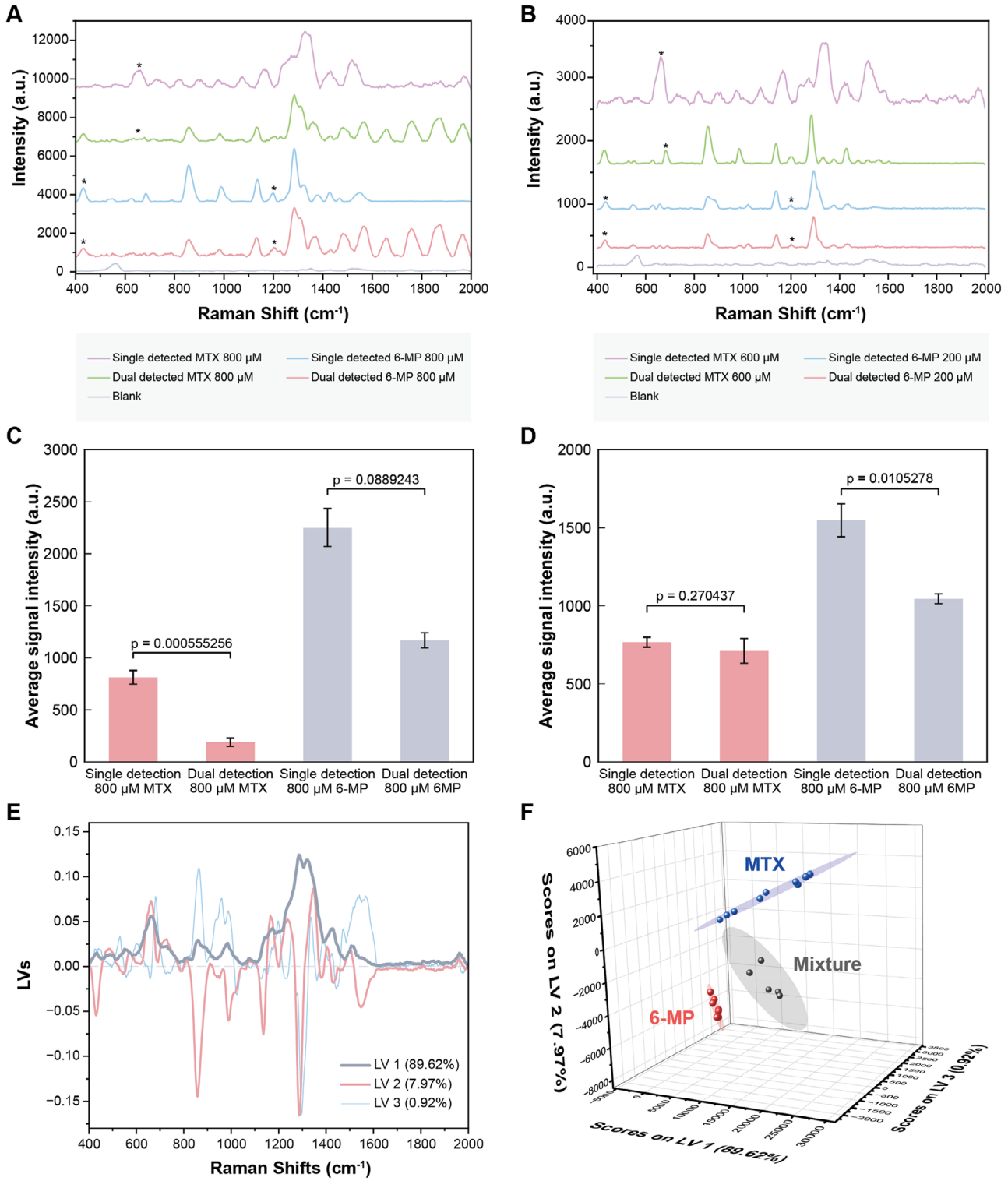


FIGURE 5 | (A) Stacked SERS spectra of dual and single detected 6-MP and MTX at 800 μM compared with blank using active Zn/Au MMs. The stars correspond to the characteristic peak at around 670 ± 5 cm^{-1} for MTX and around 440 and 1220 cm^{-1} for 6-MP, respectively. (B) Stacked SERS spectra of dual- and single-detected 200 6-MP and 600 μM MTX compared with the blank using active Zn/Au MMs. The stars correspond to the characteristic peak at around 670 ± 5 cm^{-1} for MTX and around 440 and 1220 cm^{-1} for 6-MP, respectively. Comparison of signal intensities for single and dual detection of 800 μM MTX and 800 μM 6-MP integrated with (C) Static Zn/Au MMs and (D) Active Zn/Au MMs. Corresponding error bars are depicted. p-values are shown between conditions using the two-tailed t-test with a significance level of 0.05. (E) Loading plots of PLS-DA model (LVs = 4). (F) 3D score plots of PLS-DA model for the concentration level of 600 and 800 μM MTX and 6-MP and mixture of 600 μM of each. [Correction added on June 4, 2026, after first online publication: Figure caption of Figure 5 (C) and (D) has been amended in this version.]

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ethanol to remove impurities. The purified Zn suspension was then diluted to a final concentration of 3 mg/mL. Separately, a 5 wt.% polyvinylpyrrolidone (PVP) solution was prepared, and 1 mL was spread onto a clean glass slide and left to dry at room temperature in a fume hood. The PVP film served as an adhesive layer to immobilize the Zn particles for Au-coating. Subsequently, 1 mL of the 3 mg/mL Zn solution was uniformly dispersed onto the PVP-coated glass slide and left to dry. After drying, the samples were coated with an Au layer to thicknesses of 15, 30, 37, and 45 nm using a sputter coater (Quorum Q150R, Quorum Technologies, UK), where the thickness was monitored in real time using the built-in quartz crystal microbalance (film thickness monitor). Subsequently, the glass slides were washed in ethanol to detach the MMs from the glass slide and dissolve PVP. Each washing cycle includes 20 s sonication at 25°C, followed by centrifugation at 3000 rpm for 3 min. Following the final wash, the Zn/Au MMs were resuspended and stored in DI water.

4.2 | Characterization of Zn/Au MMs

The morphology of bare Zn particles and Zn/Au MMs was characterized using a scanning electron microscope (SEM, TM3030Plus, Hitachi, Japan). Further identification and elemental analysis were performed using a FEI Quanta AFEG 250 scanning electron microscope equipped with an X-Max 50 EDS detector (Oxford Instruments, United Kingdom). The stability of the Zn/Au MMs was monitored over 15 days using SEM. SEM images were collected on days 1, 2, 5, 9, and 15. Particle stability was evaluated by counting Zn/Au MMs in 10 μL of suspension on days 1 and 15; Each drop was counted three times to obtain an average particle number. The optical properties of Zn/Au MMs with different Au thicknesses (15, 30, 37, and 45 nm) were analyzed using a Varioskan microplate reader (Thermo Scientific).

4.3 | Motion Studies

The motion behavior of Zn/Au MMs was performed using near-infrared (NIR) lasers integrated into a Leica TCS SP8 CARS microscope (Leica, Germany). All images and video sequences were recorded using Leica Application Suite X (LAS X) software, while ImageJ (version 1.54p, NIH, USA) was employed for velocity analysis. The average velocity of MMs under specific conditions was calculated by tracking the trajectories of individual MMs using the Manual Tracking plugin in ImageJ. The displacement was measured across consecutive frames and divided by the corresponding time interval to obtain the velocity of a single MM. Only MMs exhibiting stable and continuous motion were included in the analysis. For each condition, 20 MMs were tracked to determine the average velocity. First, velocity was tested at a total power of 17 mW. For other set experiments, velocity has been tested at the Pump wavelength of 800 nm. The influence of Au layer thickness (15, 30, 37, and 45 nm) was examined under two laser powers (19.2 and 43.8 mW). The effect of varying the Pump and Stokes laser powers was investigated independently, with the Pump laser fixed while the Stokes laser varied (14.7, 23.8, 31.4, and 37.1 mW) and vice versa (14.0, 24.6, 34.3, and 43.9 mW). Motion experiments in PBS were performed at pH 7.4 at 19.2 and 43.8 mW laser power. Control experiments were performed under near-UV and visible-light illumination at 405 nm (0.02 mW), 500 nm (0.02 mW), 550 nm (0.06 mW), and 600 nm (0.1 mW) using a Leica INM100 optical microscope. A summary of the velocity analysis under all experimental conditions is provided in Table S1.

4.4 | Raman and SERS Measurements

Raman and SERS measurements were conducted using a DXRxi Raman Imaging Microscope (Thermo Scientific) operated with OMNICxi software. For calibration measurements, four concentrations (200, 400, 600, and 800 μM) of MTX and 6-MP were prepared in DI water, while Zn/Au MMs were suspended in ethanol. Both static and active MMs were used. Static MMs were prepared by oven drying, whereas active MMs were prepared by laser drying under the Raman microscope. All samples were deposited on calcium fluoride (CaF_2) crystal optic slides.

Instrument calibration was performed using a silicon wafer (520.6 cm^{-1}) with the following parameters: laser power, 5 mW; exposure time, 0.1 s; two scans; image pixel size, 100 μm; and preview size, 2000 μm. For drug measurements, 50 μL of Zn/Au MMs were mixed with 50 μL of each drug concentration. For the Raman measurements, the 50× objective has been used. Raman parameters were set to a 15 mW laser power, a 2 s exposure time, one scan, an image pixel size of 2.5 μm, and 40 × 40 pixels. From each sample, four drops (10 μL each) were placed on CaF_2 slides: two for active and two for static MMs. Three Raman mappings were collected per drop.

For dual-drug measurements, two concentration sets were tested: (i) 800 μM MTX and 800 μM 6-MP and (ii) 600 μM MTX and 200 μM 6-MP. For each set, 50 μL of Zn/Au MMs were mixed with 25 μL of each drug solution. A blank control was prepared by mixing 50 μL of Zn/Au MMs with 50 μL of DI water, with one drop used for active and one for static MMs.

4.5 | Data Analysis

For the Raman maps data treatment, custom-made mapanalyzer tools have been used. The SERS mapping data were cropped between 400 and 1800 cm^{-1}, and a 9 × 10 pixel region was selected to avoid overlapping areas, resulting in three independent regions selected for analysis. Next, background correction was done by making a baseline correction using a rolling-circle filter with a radius of 100 and an ellipticity of 5. The top 5% of pixels that contributed most to the MTX band at about 670 cm^{-1} or the 6-MP peak at about 440 cm^{-1} were chosen and averaged. The theoretical limit of detection (LoD) and limit of quantification (LoQ) for both active and static Zn/Au MMs with 6-MP and MTX were calculated according to the following equations: $\text{LoD} = \frac{3\times\sigma}{S}$ and $\text{LoQ} = \frac{10\times\sigma}{S}$, where σ represents the standard deviation of the blank signal, and S denotes the slope of the calibration curve. All statistical data analysis methods were performed using the PLS toolbox (Eigenvector Research) in Matlab (2021b, MathWorks, Natick, MA, USA) and OriginPro 2022. Partial least squares- discrimination analysis (PLS-DA) was implemented as a quantitative method. Leave-one-out was utilized as a cross-validation method for the evaluation of the prediction accuracy of the models. All data points were plotted using OriginPro 2024.

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Acknowledgements

The authors would like to acknowledge the Danish National Research Foundation (DNRF122) and Villum Fonden (Grant Number. 9301) for intelligent drug delivery and sensing using microcontainers and nanomechanics (IDUN) and the Novo Nordisk Foundation (NNF22OC00722961) and the European Research Council (Grant Number. 101054945). Additionally, the Carlsberg Foundation (CF19-0379) is acknowledged for the support regarding the acquisition of the coherent anti-Stokes Raman scattering (CARS) instrument. This project has also received funding from the European Union's HORIZON TMA MSCA Doctoral Networks (HORIZON-MSCA-2023-DN-01) under Grant Agreement no: 101169012 (INTELLI-INGEST Project).

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. M. G. Vander Heiden, "Targeting Cancer Metabolism: a Therapeutic Window Opens," *Nature Reviews Drug Discovery* 10 (2011): 671–684, https://doi.org/10.1038/NRD3504;SUBJMETA.

2. Y. Han, X. L. Li, M. Zhang, J. Wang, S. Zeng, and J. Z. Min, "Potential Use of a Dried Saliva Spot (DSS) in Therapeutic Drug Monitoring and Disease Diagnosis," *Journal of Pharmaceutical Analysis* 12 (2022): 815–823, https://doi.org/10.1016/J.JPHA.2021.11.001.

3. N. O. Gomes and P. A. Raymundo-Pereira, "On-Site Therapeutic Drug Monitoring of Paracetamol Analgesic in Non-Invasively Collected Saliva for Personalized Medicine," *Small* 19 (2023): 2206753, https://doi.org/10.1002/SMLL.202206753;ISSUE:ISSUE:DOI.

4. W. Facchinatto, A. Natarajan, M. M. Gong, et al., "Multiplex Electrochemical Sensing Platforms for the Detection of Breast Cancer Biomarkers," *Frontiers in Medical Technology* 6 (2024): 1360510, https://doi.org/10.3389/FMEDT.2024.1360510/FULL.

5. K. Xu, R. Zhou, K. Takei, and M. Hong, "Toward Flexible Surface-Enhanced Raman Scattering (SERS) Sensors for Point-of-Care Diagnostics," *Advanced Science* 6 (2019): 1900925, https://doi.org/10.1002/ADVS.201900925.

6. S. E. J. Bell, G. Charron, E. Cortés, et al., "Towards Reliable and Quantitative Surface-Enhanced Raman Scattering (SERS): From Key Parameters to Good Analytical Practice," *Angewandte Chemie International Edition* 59 (2020): 5454–5462, https://doi.org/10.1002/ANIE.201908154.

7. D. Lu, Y. Chen, L. Ke, Z. Huang, Y. Lu, and J. Wang, "Bisulfite-free Detection of DNA Methylation in Early-stage Lung Cancer With a Multiplex SERS Immunoassay," *Chemical Engineering Journal* 503 (2025): 158491, https://doi.org/10.1016/j.cej.2024.158491.

8. N. Lyu, P. Potluri, V. Rajendran, and Y. Wang, "Multiplex Detection of Bacterial Pathogens by PCR/SERS Assay," *Analyst* 149 (2024): 2898–2904, https://doi.org/10.1039/D4AN00037D.

9. X. M. Qian and S. M. Nie, "Single-molecule and single-nanoparticle SERS: From Fundamental Mechanisms to Biomedical Applications," *Chemical Society Reviews* 37 (2008): 912–920, https://doi.org/10.1039/B708839F.

10. A. Thakur, R. Singh, V. Yadav, S. Siddhanta, and K. Jayaramulu, "Advancing SERS Applications of 2D Materials through the Interplay of Rational Design and Structure-Property Relationships," *Small Methods* 9 (2025): 2402056, https://doi.org/10.1002/SMTD.202402056.

11. X. Liang, N. Li, R. Zhang, et al., "Carbon-based SERS Biosensor: From Substrate Design to Sensing and Bioapplication," *NPG Asia Mater* 13 (2021): 1–36, https://doi.org/10.1038/S41427-020-00278-5;TECHMETA.

12. O. Volochanskyi, M. Švecová, V. Bartůněk, and V. Prokopec, "Electroless Deposition via Galvanic Displacement as a Simple Way for the Preparation of Silver, Gold, and Copper SERS-active Substrates," *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 616 (2021): 126310, https://doi.org/10.1016/J.COLSURFA.2021.126310.

13. G. M. Das, S. Managò, M. Mangini, and A. C. De Luca, "Biosensing Using SERS Active Gold Nanostructures," *Nanomaterials* 11 (2021): 2679, https://doi.org/10.3390/NANO11102679.

14. J. H. Aheto, X. Huang, X. Tian, X. Zhang, W. Zhang, and S. Yu, "Activated Carbon@Silver Nanoparticles Conjugates as SERS Substrate for Capturing Malathion Analyte Molecules for SERS Detection," *Journal of Food Safety* 43 (2023): 13072, https://doi.org/10.1111/JFS.13072.

15. I. B. Becerril-Castro, V. Salgueiriño, M. A. Correa-Duarte, and R. A. Alvarez-Puebla, "Nano/Micromotor-Driven SERS for Highly Sensitive and Spatially Controlled Sensing," *Advanced Functional Materials* 34 (2024): 2314084, https://doi.org/10.1002/ADFM.202314084.

16. G. Soufi, T. Maric, I. Badillo-Ramírez, et al., "SERS-Based Nano- and Microsystems Toward Biomedical Applications," *Small* (2026): 73304, https://doi.org/10.1002/SMLL.73304;PAGE:STRING:ARTICLE/CHAPTER.

17. X. Ju, C. Chen, C. M. Oral, et al., "Technology Roadmap of Micro/Nanorobots," *ACS Nano* 19 (2025): 24174–24334, https://doi.org/10.1021/ACSNANO.5C03911.

18. S. Dutta, S. Noh, R. S. Gual, et al., "Recent Developments in Metallic Degradable Micromotors for Biomedical and Environmental Remediation Applications," *Nano-Micro Letters* 16 (2024): 41, https://doi.org/10.1007/S40820-023-01259-3.

19. J. Li, X. He, H. Jiang, Y. Xing, B. Fu, and C. Hu, "Enhanced and Robust Directional Propulsion of Light-Activated Janus Micromotors by Magnetic Spinning and the Magnus Effect," *ACS Applied Materials Interfaces* 14 (2022): 36027–36037, https://doi.org/10.1021/ACSAMI.2C08464/SUPPL_FILE/AM2C08464_SI_006.MP4.

20. S. S. Das, S. Erez, E. Karshalev, Y. Wu, J. Wang, and G. Yossifon, "Switching from Chemical to Electrical Micromotor Propulsion Across a Gradient of Gastric Fluid via Magnetic Rolling," *ACS Applied Materials Interfaces* 14 (2022): 30290–30298, https://doi.org/10.1021/ACSAMI.2C02605/SUPPL_FILE/AM2C02605_SI_007.MP4.

21. A. Somasundar, A. Sen, A. Somasundar, and A. Sen, "Chemically Propelled Nano and Micromotors in the Body: Quo Vadis?," *Small* 17 (2021): 2007102, https://doi.org/10.1002/SMLL.202007102.

22. Q. Guo, H. Wang, H. Hao, C. Zhang, X. J. Liang, and D. Liu, "Micro/Nanomotor Development towards Enhanced Cancer Therapy," *Journal of Materials Chemistry B* 13 (2025): 9698–9719, https://doi.org/10.1039/D5TB01094B.

23. N. Ruiz-González, D. Esporrín-Ubieto, I. D. Kim, J. Wang, and S. Sánchez, "Micro- and Nanomotors: Engineered Tools for Targeted and Efficient Biomedicine," *ACS Nano* 19 (2025): 8411–8432, https://doi.org/10.1021/ACSNANO.4C12726.

24. T. Maric, Y. Tanaka, Z. Li, et al., "Micro/Nanomotors for Biofilm Remediation: Nanoarchitectonic Breakthroughs with a Focus on Regulatory and Clinical Translation Challenges," *Biomaterials* 326 (2025): 123715.

25. D. Danilovic, T. Maric, D. K. Bozanic, et al., "High-speed NIR Light-driven Nanomotors Prepared via Hybridization of TiO2 and Janus Ag-Ag2S Nanoparticles," *Nano Research* 18 (2025): 94907505, https://doi.org/10.26599/NR.2025.94907505.

26. I. B. Becerril-Castro, V. Salgueiriño, M. A. Correa-Duarte, and A.-P. RA, "Nano/Micromotor-Driven SERS for Highly Sensitive and Spatially Controlled Sensing," *Advanced Functional Materials* 34 (2024): 2314084, https://doi.org/10.1002/ADFM.202314084.

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

27. C. Li, W. Zhang, K. Zheng, and J. Guo, "SERS-Active Micro/Nanomachines for Biosensing," *Biosensors* 15 (2025): 115, https://doi.org/10.3390/BIOS15020115.

28. Z. Geng, G. Wang, B. Gu, et al., "Magnetically Driven Nanomotors for the Selective Extraction and Highly Sensitive SERS Analysis of Cationic Dyes," *Separation and Purification Technology* 365 (2025): 132678, https://doi.org/10.1016/J.SEPPUR.2025.132678.

29. S. Zhang, G. Yin, Z. Wang, et al., "Silicon-based Tubular Micromotor with SERS Traceability and Magnetic–Thermal Dual Responsiveness," *Nano Biomedicine and Engineering* 17 (2025): 263–276, https://doi.org/10.26599/NBE.2025.9290118.

30. X. Fan, X. Zhao, X. Tang, et al., "High-specificity SERS Sensing With Magnet-powered Hierarchically Structured Micromotors," *Optics Letters* 49 (2024): 7106–7109, https://doi.org/10.1364/OL.543066.

31. T. Yang, J. Zhou, Y. Wang, et al., "Magnetic Micromotors with Spiky Gold Nanoshells as SERS Sensors for Thiram and Bacteria Detection," *Small* 20 (2024): 2405193, https://doi.org/10.1002/SMLL.202405193.

32. Y. Xiao, L. Lan, Z. Ni, et al., "MXene-Based Micromotors: Active Molecular Enrichment and Selective Raman Enhancement," *Journal of Physical Chemistry Letters* 15 (2024): 12535–12542, https://doi.org/10.1021/ACS.JPCLETT.4C03017/SUPPL_FILE/JZ4C03017_SI_002.MP4.

33. X. Liu, Y. Wang, and Y. Peng, "Urease-Powered Micromotors with Spatially Selective Distribution of Enzymes for Capturing and Sensing Exosomes," *ACS Nano* 17 (2023): 24343–24354, https://doi.org/10.1021/ACSNANO.3C10405/SUPPL_FILE/NN3C10405_SI_007.AVI.

34. S. Chen, X. Sun, M. Fu, et al., "Dual-source Powered Nanomotor With Integrated Functions for Cancer Photo-theranostics," *Biomaterials* 288 (2022): 121744, https://doi.org/10.1016/J.BIOMATERIALS.2022.121744.

35. Y. Chen, R. Pan, Y. Wang, et al., "Carbon Helical Nanorobots Capable of Cell Membrane Penetration for Single Cell Targeted SERS Bio-Sensing and Photothermal Cancer Therapy," *Advanced Functional Materials* 32 (2022): 2200600, https://doi.org/10.1002/ADFM.202200600.

36. S. Liu, D. Xu, J. Chen, N. Peng, and T. Ma, "Nanozymatic Magnetic Nanomotors for Enhancing Photothermal Therapy and Targeting Intracellular SERS Sensing," *Nanoscale* 15 (2023): 12944–12953, https://doi.org/10.1039/D3NR02739B.

37. T. Maric, M. Z. M. Nasir, M. Budanovic, O. Alduhaish, R. D. Webster, and M. Pumera, "Corrosion of Light Powered Pt/TiO_2 Microrobots," *Applied Materials Today* 20 (2020): 100659, https://doi.org/10.1016/J.APMT.2020.100659.

38. T. Maric, M. Z. M. Nasir, Y. Wang, B. Khezri, and M. Pumera, "Corrosion due to Ageing Influences the Performance of Tubular Platinum Microrobots," *Nanoscale* 10 (2018): 1322–1325, https://doi.org/10.1039/C7NR05775J.

39. C. Chen, E. Karshalev, J. Li, et al., "Transient Micromotors That Disappear When No Longer Needed," *ACS Nano* 10 (2016): 10389–10396, https://doi.org/10.1021/ACSNANO.6B06256/SUPPL_FILE/NN6B06256_SI_006.AVI.

40. T. Maric, V. Adamakis, Z. Zhang, et al., "Microscopic Cascading Devices for Boosting Mucus Penetration in Oral Drug Delivery—Micromotors Nesting Inside Microcontainers," *Small* 19 (2023): 2206330, https://doi.org/10.1002/SMLL.202206330.

41. T. Maric, A. Løvind, Z. Zhang, J. Geng, and A. Boisen, "Near-Infrared Light-Driven Mesoporous SiO_2/Au Nanomotors for Eradication of Pseudomonas Aeruginosa Biofilm," *Advanced Healthcare Materials* 12 (2023): 2203018, https://doi.org/10.1002/ADHM.202203018.

42. M.-F. Tsai, S.-H. G. Chang, F.-Y. Cheng, et al., "Au Nanorod Design as Light-Absorber in the First and Second Biological Near-Infrared Windows for in Vivo Photothermal Therapy," *ACS Nano* 7 (2013): 5330–5342, https://doi.org/10.1021/NN401187C.

43. C. Nisula, C. Milián-Guimerá, D. K. Bozanic, et al., "Enteric Microneedles Functionalized to Photoresponsive Micromotors," *Materials & Design* 260 (2025): 115137, https://doi.org/10.1016/J.MATDES.2025.115137.

44. M. Sardo, C. Ruano, J. L. Castro, et al., "Surface-enhanced Raman Scattering of 5-fluorouracil Adsorbed on Silver Nanostructures," *Physical Chemistry Chemical Physics* 11 (2009): 7437–7443, https://doi.org/10.1039/b903823j.

45. A. O. Altun, T. Bond, W. Pronk, and H. G. Park, "Sensitive Detection of Competitive Molecular Adsorption by Surface-Enhanced Raman Spectroscopy," *Langmuir* 33 (2017): 6999–7006, https://doi.org/10.1021/ACS.LANGMUIR.7B01186.

46. L. Serioli, G. Soufi, G. Zappalà, et al., "Compact SERS detection system enabling automated assay on disc combined With advanced data analysis—A case study of methotrexate," *Sensors and Actuators B: Chemical* 419 (2024): 136375, https://doi.org/10.1016/J.SNB.2024.136375.

47. M. A. Pytlarz, G. Soufi, I. Badillo-Ramírez, et al., "Improving Meropenem Quantification in a Compact SERS-Based Centrifugal Microfluidic Platform: toward TDM of Antibiotics in ICU," *Analytical Chemistry* 97 (2025): 7888–7896, https://doi.org/10.1021/ACS.ANALCHEM.4C06902.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adfm75935-sup-0001-SuppMat.pdf.

Supporting File 2: adfm75935-sup-0002-VideoS1.mp4.

Supporting File 3: adfm75935-sup-0003-VideoS2.mp4.

Supporting File 4: adfm75935-sup-0004-VideoS3.mp4.

Supporting File 5: adfm75935-sup-0005-VideoS4.mp4

16163028, 2026, 58, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.75935 by Royal Danish Library, Wiley Online Library on [23/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License